arXiv:2502.12131v1 [cs.AI] 17 Feb 2025

Transformer Dynamics: A neuroscientific approach to interpretability of large language models

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Abstract

As artificial intelligence models have exploded in scale and capability, understanding of their internal mechanisms remains a critical challenge. Inspired by the success of dynamical systems approaches in neuroscience, here we propose a novel framework for studying computations in deep learning systems. We focus on the residual stream (RS) in transformer models, conceptualizing it as a dynamical system evolving across layers. We find that activations of individual RS units exhibit strong continuity across layers, despite the RS being a non-privileged basis. Activations in the RS accelerate and grow denser over layers, while individual units trace unstable periodic orbits. In reduced-dimensional spaces, the RS follows a curved trajectory with attractor-like dynamics in the lower layers. These insights bridge dynamical systems theory and mechanistic interpretability, establishing a foundation for a "neuroscience of AI" that combines theoretical rigor with large-scale data analysis to advance our understanding of modern neural networks.

1. Introduction

Artificial intelligence models—particularly modern deep learning systems—have scaled in both size and capability at an astonishing rate (Bahri et al., 2024). Today's large language models (LLMs), vision models, and other predictive models (e.g. recommender systems, weather prediction, navigation, etc) are operating in the real world. Yet, despite their ubiquity, we lack a comprehensive understanding of how these models work, where understanding is meant roughly as to be able to predict how a particular change to the input or to the model would affect its output in a wide range of cases. Our lack of understanding around such systems raises myriad concerns about their safety, fairness, and whether they might pose an existential risk to humanity (Amodei et al., 2016; Bengio et al., 2024).

Like the brain, deep neural networks are complex systems composed of billions of parameters interacting in highly non-linear ways. Systems with even a small number of such interacting elements can give rise to emergent behaviors that are unpredictable from a strictly bottom-up perspective (Lorenz, 1963). Consequently, it is not surprising that existing methods for investigating the workings of these models have yielded only fragmentary insights.

Current approaches in mechanistic interpretability often focus on identifying discrete circuits within neural networks—sub-networks or groups of neurons that implement particular functions (Heimersheim & Nanda, 2024; Singh et al., 2024). While these circuit-based approaches have provided some explanatory power, they tend to mirror pitfalls of early approaches to understanding the brain. One historical example in neuroscience was the concept of "grandmother cells," which posited that the unit of representation in the brain may be individual neurons that encode highly specific concepts (like a single neuron firing selectively for one's grandmother) (Plaut & McClelland, 2010). This idea, together with sparse coding—the notion that only a small fraction of neurons are active at any one time, and that representations are distributed among populations of cells (Olshausen & Field, 2004)—led to a wave of work around sparse distributed representations as a way to explain how the brain encodes information. Yet, while artificially sparsifying model activations with sparse autoencoders (SAEs) has yielded model-specific information about which sets of activations represent monosemantic concepts, many questions about how the models compute remain unanswered (Wattenberg & Viégas, 2024).

In neuroscience, a recent promising approach to interpret neural encoding of information comes from dynamical systems (Shenoy et al., 2013; Barack & Krakauer, 2021; Vyas et al., 2020). Treating the activity of populations of neu-

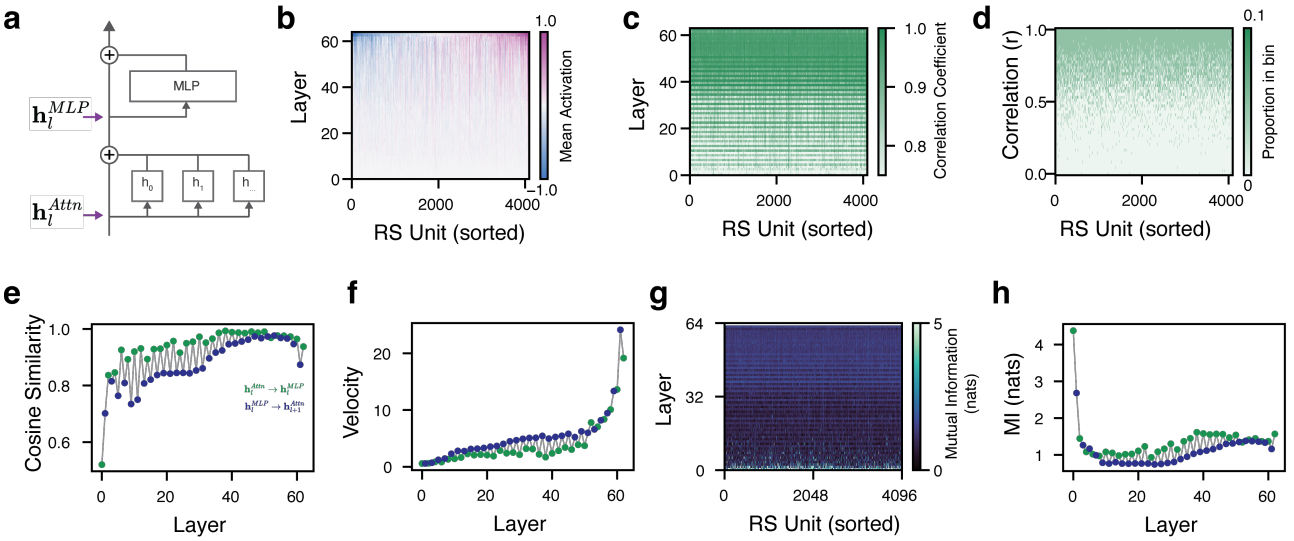

Figure 1. Transformer residual stream (RS) activations grow dense over the layers, are highly correlated among successive layers, and exhibit nonstationary dynamics. **A:** Activations of the transformer RS were captured before layernorm and the attention operation (pre-Attn) and before the MLP at each layer of Llama 3.1 8B, resulting in $\mathbf{64} \times \mathbf{4096}$. 'layers' by 'units'. Activations were analyzed at the last token position for data samples from the WIKITEXT-2-RAW-V1 dataset unless otherwise noted. **B:** Mean activations across $N = 1000$ samples. **C:** Correlations of activations for unit u between layer l and $l+1$ over data samples. For most units, correlations among successive layers increase over the layers. **D:** Histogram of correlations across layers for each unit. Despite the residual stream not having privileged basis, activations of most units are highly correlated from layer to layer. **E:** Cosine similarity among pairs of RS vectors $\mathbf{h}_l^{Attn} \to \mathbf{h}_l^{MLP}$ (green) and $\mathbf{h}_l^{MLP} \to \mathbf{h}_{l+1}^{Attn}$ (blue). **F:** Velocity V of the RS vectors. **G:** Mutual information (MI) among pairs of activations for unit u between layer l and $l+1$ over data samples. **H:** MI over the layers, averaged across units in the RS.

rons as a time-evolving dynamical system has shed light on how such populations collectively implement sensory perception, compute cognitive variables, and produce behavior. Instead of aiming to explain the representations of single neurons—or even snapshots in time of the activities of populations—these dynamical approaches aim to understand how network-wide activity evolves over time to generate complex outputs. For example, in the motor system, preparatory activity appears to guide the population to an appropriate pre-movement state; and in some cases these states are attractors that are robust to noise (Inagaki et al., 2019). Dynamical approaches have yielded insights into the computations in recurrent artificial networks as well (Maheswaranathan et al., 2019).

While transformers do not have inherent time-evolving dynamics like recurrent networks, some have examined their activations and treated them as dynamically-evolving systems (Geshkovski et al., 2024; Lu et al., 2019; Hosseini & Fedorenko, 2023; Lawson et al., 2024). Specifically, the residual stream, which is updated linearly after each layer's attention and MLP operations, can be considered as a dynamic system that evolves over the layers. Lu et al. proposed that the transformer residual stream be considered as an ordinary differential equation (ODE) of multiple particles moving through space (i.e. across layers) and influenced by convection (external forces) and diffusion (internal forces among particles); their main contribution was proposing the Strang-Marchuk splitting scheme to replace Euler's method in approximating the ODE. Geshkovski et al. similarly treat the transformer as interacting particle systems, with dynamically-interacting particles (i.e. tokens) described as flows of probability measures on the unit sphere.

This work aims to re-envision the study of mechanistic interpretability (MechInterp) through the lens of dynamical systems, inspired by this approach's success in neuroscience and driven by that field's integration of theory and large-scale data analysis. As such, we would like to term this new subset of MechInterp "the neuroscience of AI". Our key contributions are as follows:

1. We demonstrate that individual units in the residual stream maintain strong correlations across layers, revealing an unexpected continuity despite the RS not being a privileged basis.

2. We characterize the evolution of the residual stream, showing that it systematically accelerates and grows denser as information progresses through the network's layers.

3. We identify a sharp decrease in mutual information during early layers, suggesting a fundamental transformation in how the network processes information.

4. We discover that individual residual stream units trace unstable periodic orbits in phase space, indicating structured computational patterns at the unit level.

5. We show that representations in the residual stream follow self-correcting curved trajectories in reduced dimensional space, with attractor-like dynamics in the lower layers.

2. Methods

2.1. Data

Given a corpus of text sequences from WikiText-2, we first filter the dataset $\mathcal{D}$ to include only sequences s with length constraints:

$$\mathcal{D}_{\text{filtered}} = \{s \in \mathcal{D} \mid l_{\text{min}} < |s| < l_{\text{max}}\}$$

where $l_{\text{min}} = 100$ and $l_{\text{max}} = 500$ characters. For each sequence s, we obtain its tokenized representation:

$$\mathbf{x} = [t_0, t_1, ..., t_n] = \text{tokenize}(s)$$

where t_0 is the beginning-of-sequence (BOS) token. For the shuffled condition, we create a permuted sequence $\mathbf{x}'$ while preserving the BOS token:

$$\mathbf{x}' = [t_0, t_{\pi(1)}, t_{\pi(2)}, ..., t_{\pi(n)}]$$

where π is a random permutation of indices $\{1, ..., n\}$.

For each sequence (original and shuffled), we collect activations at two key points in each transformer layer $l \in \{1, ..., L\}$: Pre-attention normalization ($\mathbf{h}_l^{Attn}$) and Pre-MLP normalization ($\mathbf{h}_l^{MLP}$).

These constitute the activations of the residual stream, $\mathcal{RS}$, with $\mathbf{h}_l^{Attn}$ and $\mathbf{h}_l^{MLP}$ interleaved to make up $2L$ effective 'layers'.

We focused on the representation at the last token only, for each activation extracting:

$$\mathbf{h}_l \in \mathbb{R}^{B \times D}$$

where B is the batch size and D is the model dimension.

Altogether the extracted activations corresponded to:

$$\mathcal{RS} \in \mathbb{R}^{B \times 2L \times D}$$

For experiments in this paper, we used Llama 3.1 8B, where L = 32 and D = 4096 (Fig1A).

Since activations in the RS do not correspond to individual artificial neurons, we term each dimension in D a 'unit,' borrowing terminology from neuroscience to indicate the recording of the activation of a single element in the stream. We will thus refer to the dynamics of units in the RS as they unfold over the layers.

2.2. Transformer Residual Stream Activations

Mean activations across batches for each unit and layer were calculated in the following manner:

$$\bar{\mathbf{h}}_l^u = \frac{1}{B} \sum_{b=1}^{B} \mathbf{h}_l^{i,b}$$

Mean activations across $N = 1000$ data batches were sorted by the mean activation at the last layer:

$$\pi = \text{argsort}(\bar{\mathbf{h}}^{2L})$$

2.3. Correlations and Cosine Similarity in the Residual Stream

For each unit u, we computed the Pearson correlation coefficient between its activations at different layers across all samples:

$$r_{l,l+1}^u = \frac{\text{cov}(\mathbf{h}_l^i, \mathbf{h}_{l+1}^i)}{\sigma \mathbf{h}_l^i \sigma \mathbf{h}_{l+1}^i}$$

where $\mathbf{h}_{[i]}^l \in \mathbb{R}^B$ represents the activations of unit i at layer l across all samples.

For each unit, we analyzed the distribution of correlations across all layer pairs:

$$\mathcal{C}^i = \{r_{l,m}^i : l, m \in \{1, ..., 2L\}, l < m\}$$

The distribution was binned into intervals $[0, 1]$ to create a density plot.

Cosine similarity between consecutive layers was computed as:

$$\text{CS}_l = \frac{\mathbf{h}_l \cdot \mathbf{h}_{l+1}}{\|\mathbf{h}_l\| \|\mathbf{h}_{l+1}\|}$$

For this and other analyses, we treated the interleaved pre-attention and pre-MLP activations as layers, highlighting the two transition types: pre-attention to pre-MLP within the same layer ($\mathbf{h}_l^{Attn} \rightarrow \mathbf{h}_l^{MLP}$) and pre-MLP to pre-attention of the subsequent layer ($\mathbf{h}_l^{MLP} \rightarrow \mathbf{h}_{l+1}^{Attn}$).

2.4. Velocity

To understand how the dynamics of the residual stream change over the layers, for each layer, we calculated the magnitude of the velocity of the residual stream representation:

$$\|V_l\| = \|\mathbf{h}_{l+1} - \mathbf{h}_l\|_2$$

2.5. Mutual Information

To understand how information is processed through the layers, we analyzed the mutual information (MI) of units between layers in the residual stream. For each pair of